

MATHEMATICS OF MUSIC

Valuation Scheme - CIE 1

Course Code: MA375TGR | Semester VII | Date: 07-10-2025

Department of Mathematics

QUIZ SECTION (10 Marks)

Q1. Define frequency and state its unit of measurement [1 Mark]

Solution Steps	Marks
Frequency is the number of complete oscillations/cycles per unit time. Unit: Hertz (Hz)	1

Q2. What is the frequency ratio of an octave? [1 Mark]

Solution Steps	Marks
Frequency ratio of an octave = 2:1	1

Q3. Sound level in decibels [2 Marks]

Given: $I = 10^{-6} \text{ W/m}^2$

Solution Steps	Marks
Formula: $L = 10 \log_{10}(I/I_0)$, where $I_0 = 10^{-12} \text{ W/m}^2$	1
$L = 10 \log_{10}(10^{-6}/10^{-12}) = 10 \log_{10}(10^6) = 10 \times 6 = 60 \text{ dB}$	1

Q4. State the differences between Shruti and Swara [2 Marks]

Solution Steps	Marks
Shruti: Microtonal intervals; 22 shrutis per octave in Indian music	1
Swara: Musical notes used in practice; 7 main swaras (Sa, Ri, Ga, Ma, Pa, Dha, Ni)	1

Q5. Perfect fourth ratio in Just Intonation and cents [2 Marks]

Solution Steps	Marks
Perfect fourth ratio = 4:3	1
Cents = $1200 \times \log_2(4/3) = 498.04 \text{ cents}$	1

Q6. Harmonics with example [2 Marks]

Solution Steps	Marks
Harmonics are integer multiples of the fundamental frequency	1
For $f_0 = 100 \text{ Hz}$: 1st harmonic = 100 Hz, 2nd = 200 Hz, 3rd = 300 Hz, 4th = 400 Hz	1

TEST SECTION (50 Marks)

Q1(a). Sound wave analysis [6 Marks]

Given: $p(t) = 0.2 \sin(2\pi \times 880t)$, $v = 343 \text{ m/s}$

Solution Steps	Marks
(i) From equation $p(t) = A \sin(2\pi ft)$, frequency $f = 880 \text{ Hz}$	2
(ii) Period $T = 1/f = 1/880 = 0.001136 \text{ s} = 1.14 \text{ ms}$	2
(iii) Wavelength $\lambda = v/f = 343/880 = 0.3898 \text{ m} \approx 39 \text{ cm}$	1
(iv) Time for 50 oscillations $= 50T = 50 \times 0.001136 = 0.0568 \text{ s} \approx 56.8 \text{ ms}$	1

Q1(b). Frequency and pitch perception [4 Marks]

Solution Steps	Marks
Frequency is physical property (Hz); pitch is perceptual/psychological response	1
Higher frequency $\rightarrow$ higher perceived pitch (generally proportional)	1
Pitch perception is logarithmic because:	2
- Equal pitch intervals correspond to equal frequency ratios (not differences)	
- Doubling frequency produces one octave (constant perceptual interval)	
- Weber-Fechner law: sensation proportional to logarithm of stimulus	

Q2(a). Swaras in Indian classical music [5 Marks]

Solution Steps	Marks
Carnatic system (7 basic + 5 variants = 12 swaras): Sa, Ri ₁ , Ri ₂ , Ga ₁ , Ga ₂ , Ma ₁ , Ma ₂ , Pa, Dha ₁ , Dha ₂ , Ni ₁ , Ni ₂	3
Hindustani system (7 basic + 5 variants = 12 swaras): Sa, Re(komal), Re, Ga(komal), Ga, Ma, Ma(tivra), Pa, Dha(komal), Dha, Ni(komal), Ni	2

Q2(b). Swara frequencies using Just Intonation [5 Marks]

Given: Sa = 240 Hz

Solution Steps	Marks
Sa: $1:1 \rightarrow 240 \times 1/1 = 240 \text{ Hz}$	1
Ri: $9:8 \rightarrow 240 \times 9/8 = 270 \text{ Hz}$	1
Ga: $5:4 \rightarrow 240 \times 5/4 = 300 \text{ Hz}$	1
Pa: $3:2 \rightarrow 240 \times 3/2 = 360 \text{ Hz}$	1
Ni: $15:8 \rightarrow 240 \times 15/8 = 450 \text{ Hz}$, Verification: $Sa' = 2 \times 240 = 480 \text{ Hz}$	1

Q3. Pythagorean tuning - C major scale [10 Marks]

Starting from C, ratios and cents for all notes:

Pythagorean tuning is generated by stacking perfect fifths (3:2) and reducing by octaves

Solution Steps	Marks
C (Tonic): $1:1 = 1.0000$, Cents $= 1200 \times \log_2(1) = 0.00 \text{ cents}$	1

D (major second): $9:8 = 1.1250$, Cents = $1200 \times \log_2(9/8) = 203.91$ cents	1
E (major third): $81:64 = 1.2656$, Cents = $1200 \times \log_2(81/64) = 407.82$ cents	2
F (perfect fourth): $4:3 = 1.3333$, Cents = $1200 \times \log_2(4/3) = 498.04$ cents	2
G (perfect fifth): $3:2 = 1.5000$, Cents = $1200 \times \log_2(3/2) = 701.96$ cents	2
A (major sixth): $27:16 = 1.6875$, Cents = $1200 \times \log_2(27/16) = 905.87$ cents	1
B (major seventh): $243:128 = 1.8984$, Cents = $1200 \times \log_2(243/128) = 1109.78$ cents	1

Q4(a). Chord transposition [6 Marks]

Given: C major 7th = {0, 4, 7, 11}, transpose up perfect fourth (5 semitones)

Solution Steps	Marks
Original chord in semitones from C: {0, 4, 7, 11} = {C, E, G, B}	1
Transposition operation: add 5 semitones to each note	1
New semitones: {0+5, 4+5, 7+5, 11+5} = {5, 9, 12, 16}	2
Reduce to octave (mod 12): {5, 9, 0, 4}	1
Result: F major 7th = {F, A, C, E}	1

Q4(b). Saptak in Indian classical music [4 Marks]

Solution Steps	Marks
Saptak = octave (literally "seven notes")	1
Three main saptaks: Mandra (lower), Madhya (middle), Taar (upper)	1
Each saptak contains seven basic swaras: Sa, Ri, Ga, Ma, Pa, Dha, Ni	1
Frequency ratio between consecutive saptaks = 2:1	1

Q5. Equal temperament derivation and frequency calculations [10 Marks]

Given: $A_4 = 440$ Hz

Solution Steps	Marks
Derivation from first principles:	
In equal temperament, octave (2:1) is divided into 12 equal semitones	1
Let semitone ratio = r , then $r^{12} = 2$ (octave ratio)	1
Therefore: $r = 2^{(1/12)} = 1.05946$	1
Each semitone is separated by factor of $2^{(1/12)}$	1
Frequency calculations (Formula: $f = 440 \times 2^{(n/12)}$):	
(i) $A\#_4$: $n = +1$, $f = 440 \times 2^{(1/12)} = 466.16$ Hz	1
(ii) C_5 : $n = +3$, $f = 440 \times 2^{(3/12)} = 523.25$ Hz	1
(iii) E_5 : $n = +7$, $f = 440 \times 2^{(7/12)} = 659.26$ Hz	1
(iv) A_5 : $n = +12$, $f = 440 \times 2^{(12/12)} = 880.00$ Hz	1
(v) B_5 : $n = +14$, $f = 440 \times 2^{(14/12)} = 987.77$ Hz	1
(vi) D_6 : $n = +17$, $f = 440 \times 2^{(17/12)} = 1174.66$ Hz	1